

Experiment 2

Problem Identification and Feasibility: Software Design Principles

Loan Eligibility Prediction System

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REPO LINK	https://github.com/Dipti18-Guptq/Loan-Eligibility-Prediction-System

Experiment 2

Problem Identification and Feasibility: Software Design Principles

Aim

To identify the problem addressed by the Loan Eligibility Prediction System and prepare its problem statement, objectives, scope, and technical and operational feasibility analysis.

Objectives

- Identify the real-world problem addressed by the software project.
- Prepare a clear problem statement.
- Define project objectives and scope.
- Perform technical and operational feasibility analysis.
- Evaluate the viability of the proposed solution.

Introduction

Problem identification is the first step in software engineering. A well-defined problem statement and feasibility study ensure that the proposed software solution is practical, achievable and aligned with stakeholder requirements before development begins.

Selected Project

Project Title: Loan Eligibility Prediction System

Domain: Banking and Financial Services

Problem Statement

Loan applications require the assessment of several applicant attributes such as income, credit history, employment status, education and requested loan amount. Manual preliminary assessment can take time and may make it difficult to provide a quick and consistent initial indication of eligibility. The proposed Loan Eligibility Prediction System uses applicant information and a machine-learning based prediction process to provide a preliminary loan eligibility result in a simple and accessible manner.

Project Objectives

- Collect relevant applicant and financial information through a simple interface.
- Validate and preprocess applicant information before prediction.
- Use a machine-learning based prediction process to estimate preliminary loan eligibility.
- Display the prediction clearly so that users can understand the result.
- Use Generative AI assistance, where implemented, to explain the prediction in simple language.
- Provide a user-friendly, reliable and maintainable software solution.

Project Scope

The Loan Eligibility Prediction System covers applicant information entry, input validation, data preprocessing, eligibility prediction, result display and optional Generative AI based explanation. The project may include a frontend interface, backend/API integration and testing. Final loan sanction, banking transaction processing, account creation, disbursement and other official banking decisions are outside the scope of this project.

Technical Feasibility Analysis

Factor	Assessment	Remarks
Technology	Feasible	Python and machine-learning libraries can support data preprocessing and prediction.
Hardware	Feasible	A standard computer is sufficient for development and testing of the proposed system.
Software Tools	Feasible	VS Code/IDE, Python, Git, GitHub and UML tools can support development.
Machine Learning	Feasible	A classification model can be trained and used to generate a preliminary eligibility prediction.
Generative AI	Feasible	A GenAI component can provide supplementary natural-language explanations or assistance.
Security	Feasible	Input validation and appropriate handling of applicant information can be implemented.
Scalability	Feasible	The modular frontend/backend structure can support future enhancements.

Operational Feasibility Analysis

Factor	Assessment	Remarks
User Acceptance	High	A simple form-based interface can make the system easy to understand.
Ease of Use	High	Users only need to provide the required applicant information.
Availability	High	The software can be made available whenever the application is running.
Maintainability	High	Modular documentation and separation of components support maintenance.
Business Benefit	High	Provides a quick preliminary indication and can reduce manual preliminary assessment effort.

Analysis

The feasibility study indicates that the Loan Eligibility Prediction System is technically and operationally viable as a software project. The required development tools, machine-learning techniques and software components are suitable for implementing a preliminary prediction system.

Observation

Clearly defining the problem, objectives, scope and feasibility provides a strong foundation for subsequent software design, development and testing activities.

Result

The problem statement, objectives, project scope, and technical and operational feasibility analysis for the Loan Eligibility Prediction System were successfully prepared.